

Analytical App Development

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Prerequisites

- Fundamental programming skills
- Completion of the module SAPUI5
- Completion of the module OData
- Recommended: Completion of the module SAP Fiori

Agenda

- I. Motivation
- II. UI Controls for Simple Charts
 - a. Chart Container
 - b. Chart Types
- III. Smart UI Controls
- IV. Mockservers

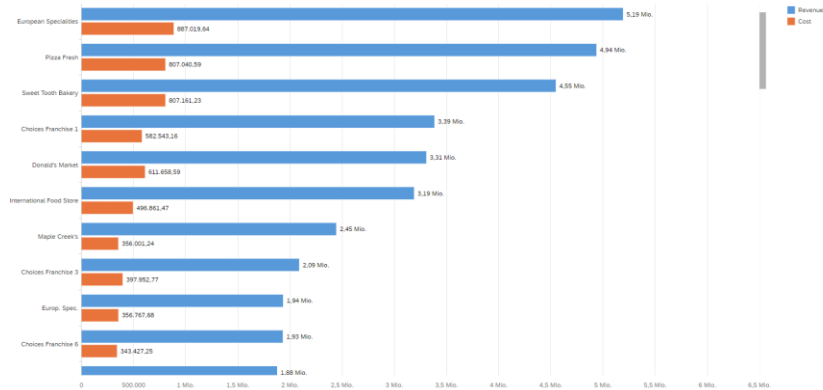
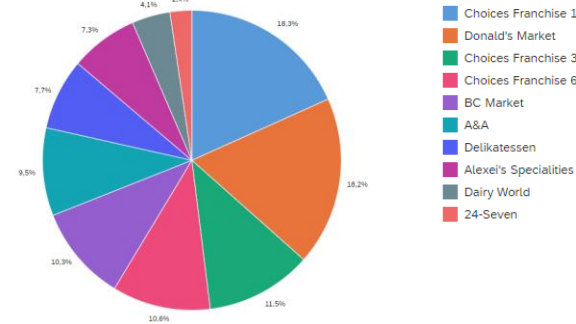
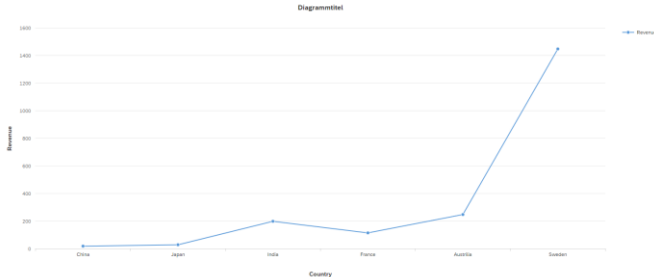
I. Motivation

- Dashboards represent a common way of information visualization
- Cover aggregated information needed usually by middle and upper management
- Involve real-time data (KPIs, sensor data, ...)
- Roll-up and drill-down functionality as must-have for multidimensional data (e.g. sales orders)

II. UI Controls for Simple Charts

Viz-Library Charts

- Displays data from data models (JSON or OData)
- Supports different chart types: line, column, pie, heatmap...
- Simple implementation – minimal need for frontend programming if data arrives in a correct form



Viz-Library Charts – API

- `setVizProperties()` – configures texts for title, category axis, and value axis
- `setDataset()` – performs data binding, requires a special Object, `FlattenedDataset`, as a parameter

```
oVizFrame.setVizProperties({
  title: {
    visible: true,
    text: title
  },
  valueAxis: {
    title: {
      visible: true,
      text: "Revenue"
    }
  },
  categoryAxis: {
    title: {
      visible: true,
      text: "Round"
    }
  }
});
```

```
var oDataset = new sap.viz.ui5.data.FlattenedDataset({
  dimensions: [{
    name: 'Roundid',
    value: "{Roundid}"
  }],
  measures: measuresRev,
  data: {
    path: "displayModel>/statistics"
  }
});
oVizFrame.setDataset(oDataset);
```

- `addFeed()` – adds a feed to a chart. `AxisLabels` and `PrimaryValues` are mandatory.

```
var feedPrimaryValues = new sap.viz.ui5.controls.common.feeds.FeedItem({
  'uid': "primaryValues",
  'type': "Measure",
  'values': valuesRev
});
oVizFrame.addFeed(feedPrimaryValues);
```

```
var feedAxisLabels = new sap.viz.ui5.controls.common.feeds.FeedItem({
  'uid': "axisLabels",
  'type': "Dimension",
  'values': ["Roundid"]
});
oVizFrame.addFeed(feedAxisLabels);
```

II. UI Controls for Simple Charts

Chart Container

- Provides functions for direct interaction with the chart
- Basic functions
 - zooming
 - enable/disable legend
 - full-screen mode
- Extendable with customized buttons and other interface elements according to your needs
- Charts need to be placed in the container

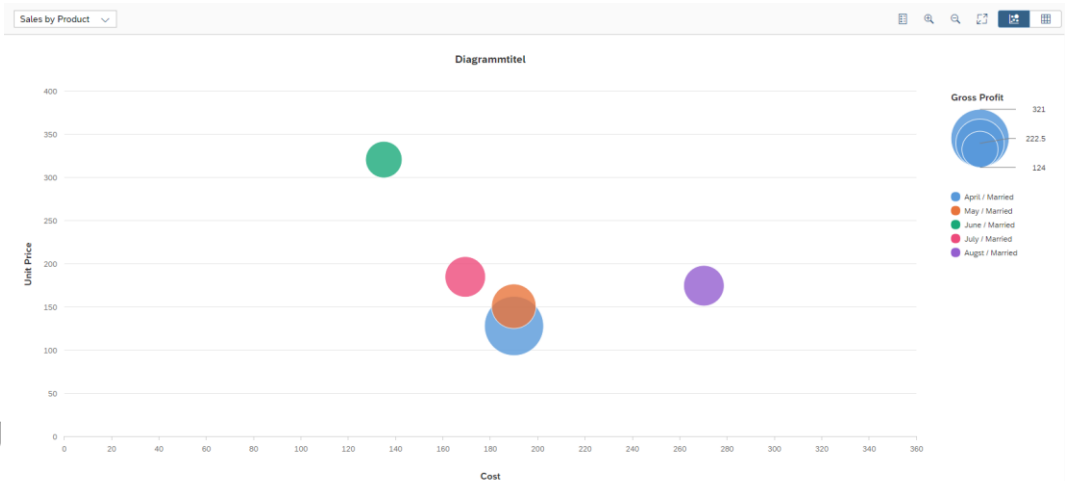


Chart Container – API

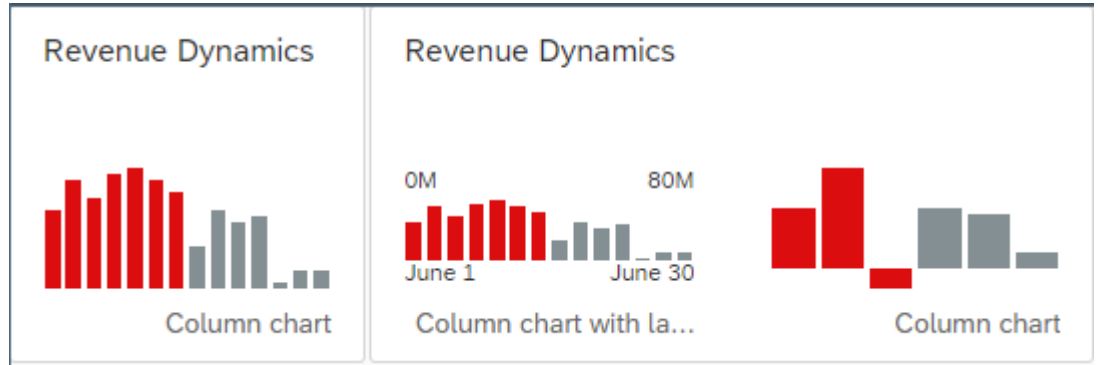
- Instantiated via constructor
- Constructor has an object of a type `ChartContainerContent` as a parameter
- `ChartContainerContent` instantiated with icon (path), title (String), and content (array of chart objects)
- Every standard container button has its own visibility setter
- Is injected into another control (e.g. as `FlexContent` of a `FixFlex`)

```
var oVizFrame = this.getView().byId("revenuePerCarriidPie");
var oChartContainerContent = new ChartContainerContent({
    icon : "sap-icon://pie-chart",
    title : "Revenue per Airline",
    content : [ oVizFrame ]
});
var oChartContainer = new ChartContainer({
    content : [ oChartContainerContent ]
});
oChartContainer.setShowFullScreen(true);
oChartContainer.setAutoAdjustHeight(true);
oChartContainer.setShowZoom(false);
this.getView().byId('chartContainer').setFlexContent(oChartContainer);
```

II. UI Controls for Simple Charts

Micro Chart

- Small layout size
- Supports all usual chart types, like bar, column, pie, etc. *
- Usually used to depict a single KPI
- Integrable into Fiori Tiles
- No interaction with single chart elements possible
 - *however the chart as a whole may react to clicks (if not integrated into a tile)*

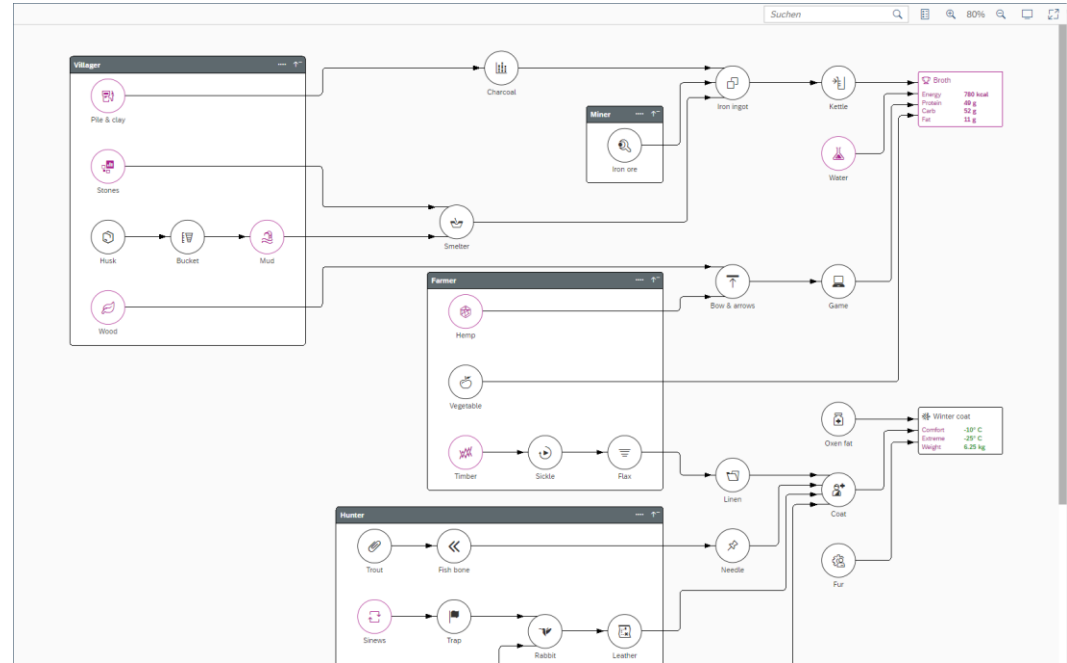


* <https://experience.sap.com/fiori-design-web/chart-types/>

II. UI Controls for Simple Charts

Network Graph

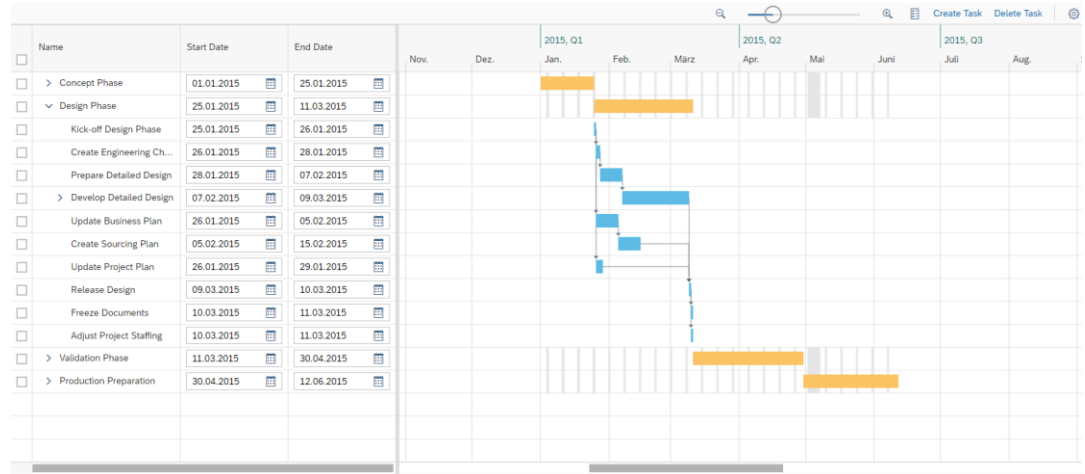
- Structural illustration of nodes and connecting lines
 - Good example – workflow visualization (on the screenshot)
- Entire layout design customizable
- Child nodes can be expanded or collapsed
- Group assignment of multiple nodes
- Minimal frontend programming necessary



III. Smart UI Controls

Gantt Chart

- Illustration of a project plan (incl. milestones)
- Bounded to a table and/or a tree map
- Self-adapting chart according to expanded nodes of an underlying tree map
- Very complex control
 - Requires high programming effort on the frontend side



Now you know the basics about the simple charts in UI5.

To consolidate your knowledge, you can do tasks 1 – 4 of the *Analytical Apps Development* exercise.

Smart Field / Smart Form

- Extended input help (not just simple values, but various search possibilities)
- Provides the user with the input help like he is used to when working with SAP GUI
- Special case: *currency fields*
 - Currency that corresponds to a selected value (e.g. flight price) is loaded from the database without the necessity to select it manually
- Usually grouped into smart forms

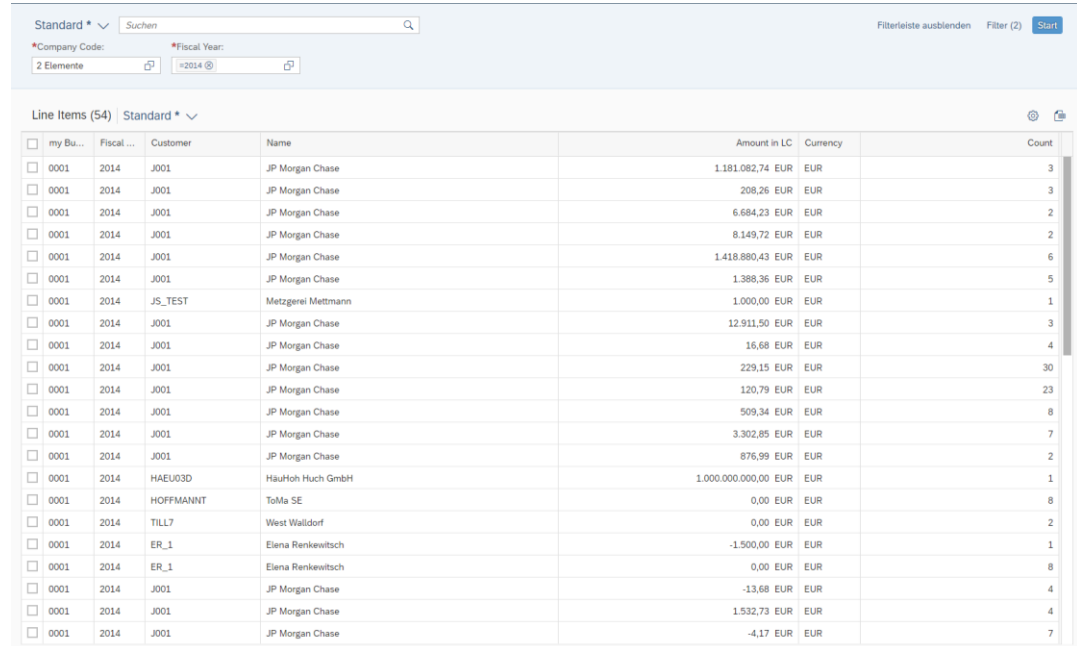
The screenshot displays the 'Company Code' smart field interface. At the top, there are two tabs: 'AUS LISTE AUSWAHLEN' (selected) and 'BEDINGUNGEN DEFINIEREN'. Below the tabs is a search bar with the placeholder 'Suchen' and a magnifying glass icon. To the right of the search bar are two links: 'Erweiterte Suche ausblenden' and a 'Start' button. Below the search bar are four input fields: 'Company Code:' with the value '*001*', 'Company Name:', 'City:', and 'Currency:'. Each field has a small icon to its right. Below these fields is a table titled 'Elemente'. The table has five columns: 'Co...', 'Company Name', 'City', 'Currency', and an empty column. The table contains several rows of data. The row with '0014' and 'test company' is selected, indicated by a blue background. Below the table is a section titled 'Ausgewählte Elemente (1)' which contains a single entry: 'test company (0014)'. At the bottom right of the interface are two buttons: 'OK' and 'Abbrechen'.

Co...	Company Name	City	Currency	
☐ 0001	SAP A.G.test	Walldorf	EUR	
☐ 0011	SAP A.G.	Walldorf	EUR	
☒ 0014	test company		EUR	
☐ 0015	test company		EUR	
☐ C001	Tools Inc. Europe	Walldorf	EUR	
☐ U001	Tools Inc. Europe	Walldorf	EUR	

III. Smart UI Controls

Smart Table

- Illustrates data as table
- Enables data sorting and filtering
- Various customization possibilities (e.g. editable cells)
- Allows creation of custom layouts (comparable with ALV grids)
- Combinable with smart fields in header



Standard * ▾ Suchen

*Company Code: 2 Elemente *Fiscal Year: =2014

Filterleiste ausblenden Filter (2) Start

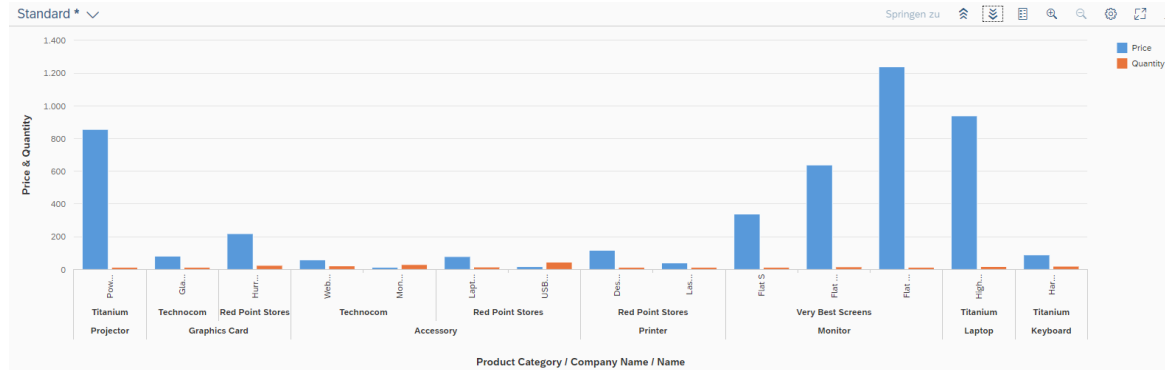
Line Items (54) Standard * ▾

☐	my Bu...	Fiscal ...	Customer	Name	Amount in LC	Currency	Count
☐	0001	2014	J001	JP Morgan Chase	1.181.082,74	EUR	3
☐	0001	2014	J001	JP Morgan Chase	208,26	EUR	3
☐	0001	2014	J001	JP Morgan Chase	6.684,23	EUR	2
☐	0001	2014	J001	JP Morgan Chase	8.149,72	EUR	2
☐	0001	2014	J001	JP Morgan Chase	1.418.880,43	EUR	6
☐	0001	2014	J001	JP Morgan Chase	1.388,36	EUR	5
☐	0001	2014	JS_TEST	Metzgerei Mettmann	1.000,00	EUR	1
☐	0001	2014	J001	JP Morgan Chase	12.911,50	EUR	3
☐	0001	2014	J001	JP Morgan Chase	16,68	EUR	4
☐	0001	2014	J001	JP Morgan Chase	229,15	EUR	30
☐	0001	2014	J001	JP Morgan Chase	120,79	EUR	23
☐	0001	2014	J001	JP Morgan Chase	509,34	EUR	8
☐	0001	2014	J001	JP Morgan Chase	3.302,85	EUR	7
☐	0001	2014	J001	JP Morgan Chase	876,99	EUR	2
☐	0001	2014	HAEU03D	Häufelh Huch GmbH	1.000.000.000,00	EUR	1
☐	0001	2014	HOFFMANNT	ToMa SE	0,00	EUR	8
☐	0001	2014	TILL7	West Walldorf	0,00	EUR	2
☐	0001	2014	ER_1	Elena Renkewitsch	-1.500,00	EUR	1
☐	0001	2014	ER_1	Elena Renkewitsch	0,00	EUR	8
☐	0001	2014	J001	JP Morgan Chase	-13,68	EUR	4
☐	0001	2014	J001	JP Morgan Chase	1.532,73	EUR	4
☐	0001	2014	J001	JP Morgan Chase	-4,17	EUR	7

III. Smart UI Controls

Smart Chart

- Extends functions of normal chart UI elements
- No chart container necessary
 - toolbar available by default
- Multiple interactions available without further configuration
 - navigation based on graph elements
 - graph types changeable at runtime
 - data roll-up/drill-down
 - layout configuration (e.g. chart legend)



Now you know the basics about the simple charts in UI5.

To consolidate your knowledge, you can do tasks 5 – 7 of the *Analytical Apps Development* exercise.

Check your knowledge

Check your knowledge

- Is it always necessary to place a chart within a chart container? In which cases is a semantic object needed?
- Why do we need analytical apps?

Solution

Solution

- Is it always necessary to place a chart within a chart container? In which cases is a semantic object needed?
 - No, it is not. SmartCharts have their own toolbar, and the viz-Charts can be depicted without the container. It just allows more interaction with the chart.
- Why do we need analytical apps?
 - Slide 4

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